

Vision-seeded geodesic partitions with budget-certified region sizing: adaptive mesh refinement in few solves for three-dimensional bridge components

— MANUSCRIPT OUTLINE v0.1 (2026-08-28, tied to branch `cursor/vision-region-amr-gmsh-e89b`) —

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Abstract

[Draft v0; every number below is from the repository pilot runs and must be regenerated by the full pre-registered campaign before submission.] Classical adaptive finite element loops (solve–estimate–mark–refine) are asymptotically optimal but need many global solves, which is precisely why they are rarely used on three-dimensional structural components. Practising engineers instead read the drawing and a coarse stress plot, circle a handful of critical zones, assign one mesh size per zone, and finalise the mesh within one or two remeshing rounds. We formalise this workflow: a vision head places *named seeds* on orthographic views of a probe solution; seeds grow into human-like, geometry-conforming regions as multi-source geodesic Voronoi cells on the element adjacency graph; each region carries a single target size, revised by one round of multi-agent negotiation driven by *measured* regional convergence exponents, and certified by a low-dimensional particle-swarm calibration that treats the equation budget as a hard cap (closed-form budget projection, tighten-only resource-drift feedback, late-round trust region, in-place certification, best-compliant-iterate delivery). All meshes are generated by Gmsh from nodal size fields; all solves use CalculiX. On two 3D bridge component families—a bearing block under eccentric patch bearing and a deck panel under wheel-patch loading—the method reaches lower energy error than a faithfully deployed element-wise Zienkiewicz–Zhu + Dör-

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fler loop at equal solve counts $k = 2-4$ (e.g. bearing block: 0.198 vs 0.291 at $k = 3$), delivering budget-compliant meshes at 83–94% of the cap without any offline training; the classical loop overtakes only from $k^* \approx 5$ solves onward while exceeding the budget. We state this crossover explicitly and cede asymptotic optimality to Dörfler by construction.

Keywords: adaptive mesh refinement, region-based mesh sizing, vision-language models, multi-agent negotiation, particle swarm optimization, bridge components

Highlights (draft; Elsevier limit: 3–5 bullets, ≤ 85 characters each)

- Vision-seeded geodesic partitions replace element-wise marking in adaptive FEM.
- One negotiation round plus PSO certifies accuracy under a hard equation budget.
- Beats Dörfler at 2–4 solves on 3D bridge bearing and deck panel benchmarks.
- Delivered meshes use 83–94% of the budget cap; overshoot is structurally barred.
- Training-free; matches learned baselines that need hundreds of offline solves.

Submission package checklist (Elsevier)

- Manuscript PDF/source (elsarticle, **preprint** option for initial submission).
- Highlights file (above); Graphical abstract: pipeline schematic (Fig. 1 crop, $\geq 531 \times 1328$ px per spec).
- Title page with authors/ORCID; CRediT statement (§9); Declaration of competing interests; Data availability statement (§9); Funding statement.
- Referee suggestions: AMR + ML-for-meshing communities (to be listed).

Target venue and article type

- **Primary:** *Advances in Engineering Software* (full-length research paper). Fit: engineering workflow formalisation + open-source pipeline; the journal published MeshingNet3D [10], a direct baseline relative.
- **Alternates:** *Computers & Structures* (structural FE workflow); *Engineering Applications of Artificial Intelligence* (agents/LLM angle); *Computer Methods in Applied Mechanics and Engineering* (only if the method section is strengthened with a formal analysis of the certified projection); *Automation in Construction* (bridge-engineering angle).
- Estimated length: 30–35 manuscript pages, 8 figures, 6 tables, 2 appendices.

1. Introduction

Purpose: establish the narrow claim — “fewest global solves to an engineering-grade mesh under a hard resource cap”, not asymptotic optimality.

- Cost model: on 3D solid components each AMR cycle costs one global solve; 8–12 cycles is exactly why practitioners avoid adaptive FEM (§1.1 of the experiment plan).
- The human protocol: read drawing + coarse contour plot → circle a few zones → one size per zone → finalise in 1–2 remeshes. This paper is its formalisation and quantitative defence.
- Problem statement (informal): given instance, equation budget $N_{\text{eq}}^{\text{max}}$ and solve budget k_{max} , deliver a budget-compliant mesh minimising energy error, counting *every* global solve including the probe.
- Pre-registered hypotheses H1–H4 (verbatim from the plan, §7): few-solve superiority; hard-cap compliance [90%,105%]; training-free parity; honest crossover k^* .
- Contributions:

C1 Human-like partition representation: named vision seeds grown into geodesic Voronoi regions on the element adjacency graph; residual-driven region splitting; *regions are never boxes*.

- C2 Budget-certified few-solve loop: one communication round with measured exponents + (s, κ) PSO under a hard cap with closed-form projection, tighten-only drift feedback, trust region, in-place certification and best-compliant-iterate delivery semantics.
 - C3 Faithful single-mesher redeployment of four comparison methods (Dörfler, one-step local prediction, region-graph DQN, supervised size field) with pre-registered axes and machine-checkable validity gates G1–G7.
 - C4 Evidence on 3D bridge components that the loop wins at $k = 2$ –4 while explicitly exhibiting the crossover where element-wise marking prevails.
- Scope disclaimer paragraph: linear elasticity, C3D4, ZZ indicator; the claim is about *solve-count efficiency under a cap*, nothing else.

2. Related work

Purpose: four clusters, one gap. (Table 1; representative citations to be completed.)

- Known failure modes of one-step local prediction (oscillation, coarsening overshoot, invalid smoothness assumption near singular lines, pre-asymptotic indicator noise, uncertified budgets) documented in [5, 6]; each maps to a VLA counter-mechanism (§4) and an ablation (§6).
- Positioning sentence: first *closed-loop* vision-partition AMR with hard budget certification, evaluated against faithfully deployed classical and learned baselines on 3D structural components.

3. Problem setting: budgeted few-solve AMR

- Model problem: linear elasticity, energy norm; QoI = load-patch mean displacement modulus (reported separately, never mixed).
- Error metric $e_E = \sqrt{1 - U_h/U_{\text{ref}}}$; reference solutions from analytically graded size fields ($r^{0.6}$ along singular lines with floor $h_{\text{ref}}/8$; $r^{0.55}$ at corners); validity gate G3 ($U_h \leq U_{\text{ref}}$ always, zero `above_reference` records).

Table 1: Related-work map (manuscript Table 1).

Cluster	Representatives	Relation to this paper
Classical element-wise AFEM	Dörfler [3]; ZZ [2]; Babuška–Rheinboldt [1]; Li–Bettess [4]	Comparison target deployed per the sources; asymptotic optimality conceded.
RL for AMR	Yang et al. [7]; Foucart et al. [8]; ASMR [9]	RL baseline uses <i>regions</i> as graph nodes, aligned with our decision object.
Supervised size fields	MeshingNet/3D [10]; AMBER [11]	Supervised baseline: self-generated expert meshes + budget scalar.
MLLM hot-spot circling (no closed loop)	GReFEM [12]; FeaGPT [13]	They circle hot spots once, without residual revision or budget certification. <i>Gap this paper fills.</i>

- Budget semantics: equation cap $N_{\text{eq}}^{\max}$ is *hard*; element budgets converted by measured equations-per-element ratio (dimension prior 0.62 cap on coarse meshes).
- Solve accounting: probe counts as solve 1 for every method; PSO consumes only surrogate evaluations; training solves reported in a separate column (axis A4).
- Formal statement of the comparison axes A2' (error @ k solves; the main axis), A2'' (speed-up ratio), A1 (error- N_{eq} , log-log, crossover exhibited), A3 (budget compliance), A4 (end-to-end training cost).

4. Method: the vision-seeded region loop (VLA)

Manuscript figure 1: pipeline schematic (to draw); figure 2: partitions (exists, Fig. 1).

4.1. Vision heads (ablation axis AB1)

- **LLMVisionPartitioner** (headline): three orthographic rendered views of the probe field $\rightarrow$ multimodal LLM $\rightarrow$ strict-JSON named seeds (position + fineness score), instructed to emit both hot-spot seeds and “keep-coarse” field seeds; retry $\times 3$ + scripted fallback with reported fallback rate (risk register).
- **ScriptedVisionPartitioner** (reproducible stand-in / lower bound): von Mises nodal local maxima with non-maximum suppression $\rightarrow$ peak seeds; *drawing-level structural anchors* (patch edge midpoints, support strip third-points, re-entrant corners, hole rims — engineering common-sense declared as feature anchors) $\rightarrow$ anchor seeds; far-point sampling in low-field zones $\rightarrow$ coarse seeds.

4.2. Human-like partitions: seeds grown into geodesic regions

- On any mesh, the partition is the multi-source *geodesic Voronoi* decomposition of the element adjacency graph (each element joins its graph-nearest seed). Regions hug geometry: bands along support strips, rings around load patches, wrap-around at thin walls; full coverage, no background pseudo-region; *never axis-aligned boxes* (ablation AB2 quantifies the difference vs the box implementation).
- One size per region; granularity grows by *region splitting*: a region concentrating residual spawns a child seed at its peak element (≤ 2 splits/round, ≤ 14 regions total) — the formal analogue of a human “drawing a smaller circle inside”.

4.3. One communication round with measured exponents

- Region features: log size, error and element shares, von Mises level, volume share, realised mean size, remaining solves, budget occupancy.
- One round of neighbour/parent messages produces bounded log-step size revisions; the local convergence exponent q_i entering the update is *fitted from two consecutive real solves* (clipped to $[0.6, 4.5]$), not assumed smooth — the direct counter to the oscillation root cause of [5] (ablation AB9).

4.4. PSO calibration under a hard cap; certification semantics

- Two variables only: $h_i = h_i^+ \exp(s + \kappa\tau_i)$ with τ the marginal-efficiency resource-neutral transfer direction; power-law surrogate anchored at the nominal sizes of the latest real solve (mother-particle self-consistency); same fitted q_i as the communication round.
- Fitness: $400 \cdot \text{overshoot}^2 + 200 \cdot \min(e^+, 1)^2 + 10 \cdot \text{linear accuracy drive} + \text{neighbour-ratio and deviation penalties}$; best particle receives a closed-form *budget projection* (safety 0.97 regular rounds, 0.85 final round).
- Tighten-only *resource-drift* feedback: projected budget divided by realised/predicted ratio, clipped to $[1.0, 1.4]$ — the mesher’s gradation overhead can only shrink the target, never inflate it (ablation AB10).
- Late-round trust region: PSO position bound $0.8 \rightarrow 0.45$ from round 3 (ablation AB7/AB10 interaction).
- *In-place certification*: if the current solve uses $\in [88\%, 100\%]$ of the cap and the surrogate predicts $<15\%$ gain, certify on the spot and save the remaining solves (ablation AB11).
- *Delivery semantics*: among budget-compliant solved iterates, return the one with the smallest own indicator sum (return-best-iterate; the reference solution is never consulted for the pick).
- Algorithm box (manuscript Algorithm 1): probe $\rightarrow$ seed/partition $\rightarrow$ equidistribution init $\rightarrow$ communicate $\rightarrow$ PSO $\rightarrow$ solve $\rightarrow$ [fit exponents $\rightarrow$ communicate $\rightarrow$ split $\rightarrow$ PSO $\rightarrow$ solve]* $\rightarrow$ certified, $\leq k_{\max}$ solves, adaptive early stop.

5. Experimental setup

5.1. Benchmarks: 3D bridge components (headline) + 2D substrates (CI only)

- **bearing_block**: $400 \times 400 \times 120$ mm bearing plate, fixed base, eccentric top patch pressure; 4 singular patch-edge lines + clamped-perimeter reaction lines. Prototype: bearing upper plate / pedestal local bearing.
- **deck_panel**: $2400 \times 1600 \times 200$ mm deck slab on two girder support strips ($u_z = 0$), wheel patch on top; bending hot spot + strip inner edge reaction lines. Prototype: girder-bridge carriageway slab wheel check.

- Load/constraint footprints *imprinted into the OCC geometry* (`occ.fragment`) so every mesh tiles patch and strip boundaries exactly; resultant force equals $p \times A$ on any mesh (gate G7). Lesson learned in v2: without imprinting the loaded area drifts with the mesh and the energy error is non-monotone.
- Instance sampler (train 24 / test 8 + canonical, seeds locked): bearing patch 100–180 mm, eccentricity ± 70 mm, 8–16 MPa; deck wheel position $x \in [700, 1700]$, $y \in [500, 1100]$, 0.7–1.4 MPa, patch $300\text{--}500 \times 200\text{--}320$ mm. Learned methods evaluated on the test split only.
- 2D substrates (L-bracket corner singularity, plate with holes) retained for unit tests and the G1 smoke gate; main text reports 3D only.

5.2. Shared infrastructure (locked)

- Gmsh ≥ 4.11 (pilot 4.15.2), Frontal-Delaunay 2D / HXT 3D, linear CPS3/C3D4; every decision becomes a nodal size map $\rightarrow$ Lipschitz gradation ($g = 0.9$) $\rightarrow$ `setSizeCallback` remesh; *no hand-written mesh operations anywhere in the repository* (gate G2, structural).
- CalculiX ≥ 2.21 (SPOOLES); every call logged by the **FemRunner** ledger (method/stage/ N_{eq} /elements/ U /QoI/wall-clock) — gate G5.
- ZZ (1987) volume-weighted nodal recovery in the energy norm; 3-midpoint (2D) / 4-point (3D) exact quadrature; deliberately not SPR (uniform across methods).
- Budget tiers: bearing $N_{\text{eq}} \in \{4000, \mathbf{8000}, 16000\}$; deck $\{10000, \mathbf{20000}, 40000\}$ (bold = pilot tier). Randomness: PSO seed 17, RL/supervised seeds explicit, single-thread deterministic meshing.

5.3. Baseline deployment contracts (audited; manuscript Table 2)

- **Uniform ladder:** ratio $2^{1/d}$ per rung (element count doubles), stop at cap. Answers “what do DOFs alone buy”.
- **Dörfler element-wise loop:** unit = element; ZZ indicator; bulk marking $\theta = 0.5$ (minimal cardinality, unit-tested); marked element size halved $\rightarrow$ Gmsh remesh; stop on empty marking / cap / 12 rounds. Full curve recorded from cycle 0.

- **Local prediction** (one-step multi-level, the “few-solve” classical rival): ZZ remeshing + Li-Bettess optimality criterion; exponent = local theoretical rate $\eta_K \sim h^{(d+2)/2}$; per-round size-ratio clip $[1/6, 1.8]$ (tightened coarsening bound — the 3D pilot reproduced the known oscillation: 0.18→0.30 rebound under a loose bound); per-tier short runs (probe+2), budget deviation reported as-is.
- **Region-graph DQN** (RL baseline): graph nodes = partition regions (fixed per episode); 8-dim features; 2-layer GCN → per-region Q + global stop-Q; action = region size $\times 0.7$ or stop; every step is a real remesh+solve; oracle-free reward; Double DQN; 300 eps $\times$ 3 seeds (2D), 120 eps $\times$ 3 seeds (3D, scale reported honestly).
- **Supervised size field**: experts = final Dörfler-at-cap meshes on the 24 training instances; 9-dim nodal features → MLP 64×64 regressing $\log(h/h_0)$; deployment = probe + budget scalar + one remesh = 2 solves.
- Cross-reference to `docs/BASELINE_AUDIT.md`: list of defects found in the historical implementations and their fixes (one summary table in the appendix; builds reviewer trust that baselines are not straw men).

5.4. Protocol, statistics, and validity gates

- Median [IQR] over the 8 test instances; paired Wilcoxon for H1 ($\alpha = 0.05$); 3 seeds for RL/PSO.
- Figure discipline (anti-patterns banned): no mixed-refinement-atom Pareto fronts; no stitching local-prediction tiers into one convergence curve; the crossover k^* must be drawn.
- Gates G1–G7 (deployment correctness vs uniform on 2D; Gmsh sole mesher; reference validity; distinct solves; honest counting; figure policy; load consistency) — all script-checked; pilot status: all pass.

6. Results (planned; pilot evidence embedded)

Main table = manuscript Table 3 (pilot values below, to be regenerated on the test split with statistics).

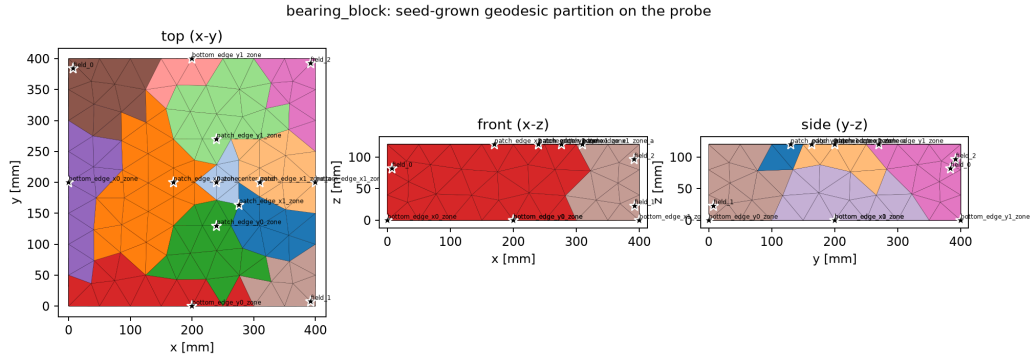
- §6.1 Error @ k solves + speed-up ratios ($A2'/A2''$): Table 2 expanded to the test split; Wilcoxon column; target claim H1 (speed-up $\geq 1.5 \times$ to reach Dörfler round-4 accuracy).

Table 2: Pilot headline: energy error e_E after the k -th global solve (canonical instances, scripted head, deterministic meshing; repository reruns). VLA delivery = best budget-compliant iterate by its own indicator.

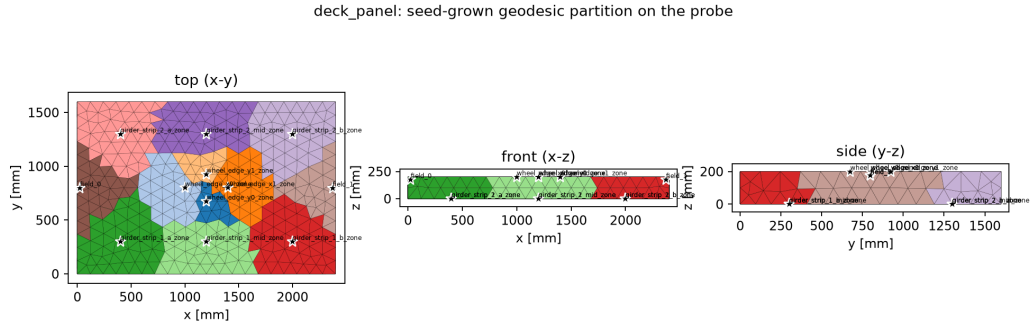
k	Bearing block (cap 8000)		Deck panel (cap 20000)	
	Dörfler	VLA	Dörfler	VLA
2	0.375	0.212	0.601	0.351
3	0.291	0.198 (cert., stop)	0.492	0.351
4	0.210	— (delivered at 3)	0.388	0.341
5	0.143*	—	0.274	—

*Dörfler round 5 exceeds the cap by 17% ($9354 > 8000$). Delivered budget use: bearing 94% (7500), deck 83% (16628).

- §6.2 Error- N_{eq} trade-off (A1): log-log with refinement atoms annotated and crossover $k^* \approx 5$ marked (Fig. 2; honesty axis H4).
- §6.3 Budget compliance (A3): $N_{\text{eq}}/\text{cap} - 1$ scatter across methods; pilot anchors: VLA 83–94%; Dörfler +17% (bearing r5), +90% (deck r6); local prediction $-26\% - +59\%$ uncertified.
- §6.4 Training cost (A4): deployment solves vs offline solves (RL $\approx 600 - 1500 \times 3$; supervised ≈ 200 ; classical/VLA = 0); canonical supplements: uniform h5 0.203@11763 (bearing), 0.326@35875 (deck); local prediction 0.179@5961 / 0.269@17552; supervised (2D demo) 0.050@8702 in 2 solves; RL 2D demo-scale plateau 0.17.
- §6.5 Ablations AB1–AB11 (one condensed table + small multiples): vision head (LLM vs scripted vs random seeds), geodesic vs box partitions, structural anchors on/off (pilot: without base-edge anchors bearing $k=4$ degrades $0.180 \rightarrow 0.21$ and overshoots budget), splitting on/off, communication on/off, PSO certification on/off, $k_{\text{max}} \in \{3..6\}$, (s, κ) vs s -only vs per-region Nelder–Mead, measured vs assumed exponents, drift feedback and safety factor sweep, in-place certification gate sweep.
- 2D substrate note (one paragraph): VLA certifies at solve 2 on both substrates (L-bracket 0.098@7704, 96% cap; plate-holes 0.049@6743, 84%) — the adaptive solve count also shortens.

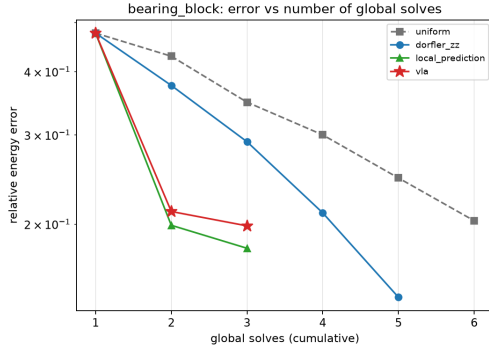


(a) Bearing block: seeds + geodesic regions.

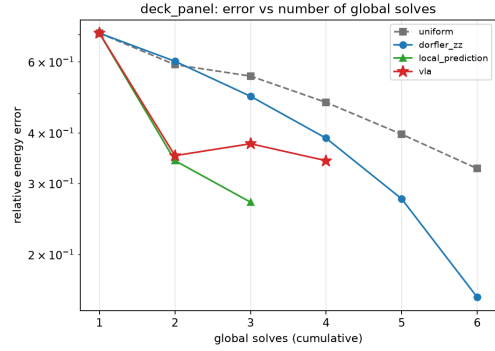


(b) Deck panel: seeds + geodesic regions.

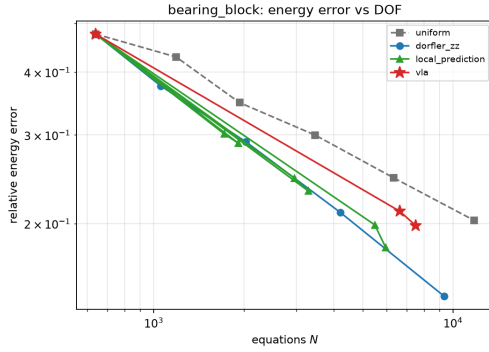
Figure 1: [Pilot evidence for manuscript Fig. 2] Vision-seeded geodesic partitions on the two 3D bridge components (three orthographic views each; colours = regions, markers = named seeds). Source: docs/evidence/.



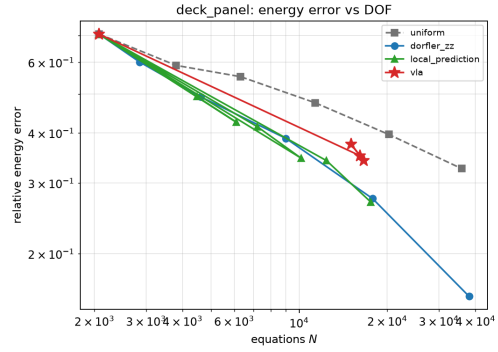
(a) Bearing block.



(b) Deck panel.



(c) Bearing block, error vs N_{eq} .



(d) Deck panel, error vs N_{eq} .

Figure 2: [Pilot evidence for manuscript Figs. 4–5] Top: main axis A2', energy error after the k -th global solve. Bottom: axis A1, error- N_{eq} log-log with differing refinement atoms annotated; the Dörfler crossover is exhibited, not hidden. Source: docs/evidence/.

7. Discussion

- Verdict subsection per pre-registered hypothesis: H1 (few-solve superiority, $k \in \{2, 3, 4\}$ + speed-up $\geq 1.5\times$), H2 (hard-cap semantics: delivered $N_{\text{eq}} \in [90\%, 105\%]$ of cap for VLA vs uncertified deviations $\geq 20\%$ for one-step prediction), H3 (training-free parity with learned baselines; total-cost accounting), H4 (crossover k^* stated: pilot ≈ 5).
- Honesty boundaries (verbatim commitments): Dörfler overtakes for $k \geq k^*$; local prediction wins some k points on raw energy error while violating budget discipline (uncertified, $-26\%/+59\%$); 3D RL and supervised results are demo-scale and labelled as such.
- Why the mechanism works (tie each gain to an ablation): measured exponents kill the oscillation root cause; regional aggregation denoises pre-asymptotic indicators; drift feedback + projection make the cap a structural invariant rather than a hope; anchors inject drawing knowledge invisible to field-only indicators.
- Cost realism: PSO consumes surrogate evaluations only; LLM head latency and fallback rate; wall-clock per solve (1–20 s on the pilot 4-core VM).

8. Limitations and future work

- Linear elasticity, C3D4 locking bias (same element family for all methods, so rankings remain meaningful; noted with G3 reference validation), single indicator (ZZ), one size per region, scripted head as vision-quality lower bound, LLM reproducibility handled by strict JSON + fallback reporting.
- Extension card (explicitly out of scope): contact/friction bearings, moving wheel-load envelopes, C3D10 verification, real bridge-drawing views for the LLM head, goal-oriented (dual-weighted) indicators.

9. Conclusions

- One-paragraph restatement of the narrow claim with final campaign numbers.

- The engineering message: a certified, training-free, few-solve regional loop formalising what analysts already do — with the classical loop’s asymptotic superiority explicitly conceded and located (k^*).

CRedit authorship contribution statement

Author 1: Conceptualization, Methodology, Software, Investigation, Writing – original draft. Author 2: Validation, Writing – review & editing, Supervision. [finalise]

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships.

Data availability

All code, locked seeds, benchmark scripts and evidence figures are in the public repository (`visionamr/`, `scripts/`, `docs/evidence/`); raw `.frd` ledgers of the full campaign will be archived (Zenodo DOI) on acceptance. Reproduction entry points: `scripts/run_benchmark.py -problem bearing_block -n-eq-budget 8000`, `scripts/run_benchmark.py -problem deck_panel -n-eq-budget 20000`, `scripts/make_figures.py`, gate check `scripts/run_smoke.py`.

Acknowledgements

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Appendix A. Locked parameters (manuscript Appendix A)

- VLA: gradation $g = 0.9$; exponent clip $[0.6, 4.5]$; PSO seed 17, position bound $0.8 \rightarrow 0.45$ (round ≥ 3); projection safety 0.97 / final 0.85; drift clip $[1.0, 1.4]$; in-place certification gate 88%/15%; splits ≤ 2 per round, ≤ 14 regions; accuracy limit = $0.25 \times$ probe indicator sum.
- Local prediction: exponent $(d + 2)/2$; ratio clip $[1/6, 1.8]$.
- DQN: 8-dim region features, 2-layer GCN, Double DQN, reward $10 \Delta \log \sqrt{\Sigma \eta^2} - 0.15/\text{step}$, over-budget -2 terminal, stop reward $2(N_{\text{eq}}/\text{cap}) - 1$.
- Supervised: 9-dim nodal features, MLP 64×64 , target $\log(h/h_0)$.
- Full table generated from `locked.json` at campaign start (step S0).

Appendix B. Reproducibility and audit map (manuscript Appendix B)

- Repository layout table (mirrors README): `visionamr/` `core`, `baselines/`, `vla/`, `scripts/`, `tests/`, CI workflow with G1 smoke gate.
- Baseline audit summary table from `docs/BASELINE_AUDIT.md` (historical defect $\rightarrow$ fix $\rightarrow$ unit test).
- Campaign step cards S0–S9 with artefact names (no calendar claims).

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